Danio rerio (Zebrafish) as a Model System for Studying Mitochondrial Disease and Dysfunction

By Kashish Gupta

Introduction:

Danio rerio, commonly referred to as the zebrafish, is a tropical freshwater fish from the Cyprindae family. It was first introduced by Streisinger and colleagues in the early 1980s as a model system. The high degree of genomic conservation between zebrafish and humans has been a major driver of its adoption, with approximately 70% of human genes having at least one clear zebrafish ortholog, compared to roughly 80% in mouse models (Steele et al., 2014). This substantial genetic overlap, combined with the ease of genetic manipulation, has made zebrafish a powerful model for studying human genetic diseases *in vivo*.

The zebrafish possess several advantages over traditional rodent models in the study of vertebrate development and disease. These include hundreds of embryos in a single clutch and optical clarity of developing embryos, which allows live imaging at the organism level. The use of tissue-specific transgenic animals can be easily generated under the control of various selected gene promoters (Broughton et al., 2001). They also have a low cost of maintenance and availability of many easily applicable genetic and genomic methods. One unique advantage is also that the zebrafish mitochondrial DNA has also 70% identity to human mitochondrial DNA and the same complement of 37 genes. Gene order, strand-specific nucleotide bias, and codon usage are conserved across vertebrate mitochondrial genomes, supporting the translational relevance of zebrafish models for human mitochondrial biology (Steele et al., 2014).

Mitochondria are found in all nucleated cells, and commonly associated with the tagline “powerhouse of the cell”, as they generate adenosine triphosphate (ATP) via oxidative phosphorylation. They have their own unique circular genome, mitochondrial DNA (mtDNA) encoding 22 transfer RNAs and 2 ribosomal RNAs, as well as 13 proteins, all of which are subunits of enzymes of the electron transport chain. They are not only involved in cellular respiration, but also involved in apoptosis, calcium storage, signaling events, and metabolism. Mitochondria are highly dynamic and mobile structures, and their function is affected by numerous factors: proteins, number, cellular localization, movement within the cell, etc. (Steele et al., 2014). When mitochondrial fusion and fission events become unbalanced or eliminated (for example, by the loss of function of the proteins required for these processes), the morphology, DNA stability, and respiratory capacity of mitochondria can be impacted negatively. Due to the advantages of using zebrafish as a model, as described above, mitochondrial diseases (MDs) that arise from mutations in both the mitochondrial genome and in nuclear genes that encode mitochondrial proteins can be explored and modeled (Pereira et al., 2024).

The zebrafish model has enabled significant advances in the study of mitochondrial disease and dysfunction. Zebrafish have been used to model disorders arising from mutations in both mtDNA and nuclear genes encoding mitochondrial proteins, including defects in oxidative phosphorylation, cardiolipin remodeling, mitochondrial dynamics, and mitophagy. Studies leveraging this model have provided insight into developmental consequences of mitochondrial dysfunction, tissue-specific vulnerability, and compensatory metabolic responses at the organismal level. Moreover, the compatibility of zebrafish with transcriptomic, proteomic, and metabolic profiling has allowed researchers to systematically characterize mitochondrial disease mechanisms and identify conserved pathways relevant to human pathology. Together, these advances establish zebrafish as a robust and translationally relevant model for elucidating the molecular basis of mitochondrial disease.

Methods:

The goal of transcriptomics is to catalogue all species of transcript, including mRNAs, non-coding RNAs and small RNAs; to determine the transcriptional structure of genes; and to quantify the changing expression levels of each transcript during development and under different conditions. To explore the manifestation of MDs due to mitochondrial dysfunction, transcriptomics, specifically RNA-seq, is widely utilized. RNA-seq is a developed approach to transcriptome profiling that uses deep-sequencing techniques.

The analysis starts with the extraction and isolation of RNA from a biological sample, such as a cell line or a frozen tissue sample. Generally, a population of RNA is converted to a library of cDNA fragments with adaptors attached to one or both ends. Each molecule, with or without amplification, is then sequenced in a high-throughput manner to obtain short sequences from one end, which is single-end sequencing, or both ends (pair-end sequencing). The analysis requires specialized computational tools that can account for the shortcomings of sequencing technologies, including the generation of sequencing errors, length biases, and fragmentation. This technology is critical to mitochondria studies within the zebrafish, as it allows a view inside the mechanisms of dysfunction and its effect on expression levels (Deshpande et al., 2023).

In zebrafish specifically, the "*Tafazzin*-deficient zebrafish display mitochondrial dysfunction, neutropenia, and metabolic defects without myopathy" by Usua et al. outlines the RNA-seq procedure. Overall, the article explores the effect on the zebrafishes' development and survival as a result of being *tafazzin*-deficient, which encodes a phospholipid-lysophospholipid transacylase located in the mitochondria inner membrane. In this study, groups of 25 larvae were collected at 5 days post-fertilization (dpf) and euthanized by immersion in an ice slurry. Each tafazzin maternal zygotic and wild-type fish was dissected for RNA extraction from the heart and whole kidney marrow, and the resulting RNA was isolated using TRIzol.

Tree pools of maternal zygotic *tafazzin* mutants and 3 pools of wild-type from the same clutch were compared, and the quality of the RNA was determined by Bioanalyzer (Agilent), the quantity measured by RT-qPCR (another method that is used frequently in mtDNA studies). Sequencing reads generated from the Illumina platform were assessed for quality and trimmed for adapter sequences using TrimGalore, a wrapper script for FastQC and cutadapt. The high quality reads were then aligned to the zebrafish reference genome and analyzed for differential expression using Cufflinks, a RNA-seq analysis package which reports the fragments per kilobase of exon per million fragments mapped for each gene. There was a significance cutoff of q-value < 0.05, and performed a gene enrichment analysis for the genes under selection. In this way, the researchers were able to profile global gene expression in whole larvae and identify transcriptional changes associated with mitochondrial dysfunction (*Tafazzin-Deficient Zebrafish Display Mitochondrial Dysfunction, Neutropenia, and Metabolic Defects without Myopathy* | *Scientific Reports*, n.d.)

Results:

This model has been used to answer essential biological questions correlating with human disease and possible areas for therapeutics.

In "Transcriptome analysis of atad3-null zebrafish embryos elucidates possible disease mechanisms" by Ezer et al., the nuclear gene *ATAD3A* was analyzed. It encodes the ATAD3A protein, which functions as a hexamer spanning the inner and outer mitochondrial membranes. It is crucial to mitochondrial dynamics and influences the structure and function of the mitochondrial inner membrane and cristae, causing elongated or fragmented mitochondria when over or under expressed, respectively. It is also implicated in mtDNA maintenance, and the loss of ATAD3A results in assimilation of free cholesterol and triglycerides in hepatocytes, promoting nonalcoholic fatty liver disease, for example. In neurons, the protein's oligomerization induces cholesterol accumulation by inhibiting CYP46A1. All this to say, ATAD3A is an essential protein in mitochondrial functions, and since the zebrafish ortholog gene *atad3* shares about 8-% similarity with *ATAD3A,* it was used to start to bridge the knowledge gap regarding the mechanisms of disease in humans.

CRISPR/Cas9 genome editing was used to generate a knockout model of *atad3*. Imaging analysis, quantification of absolute mtDNA amount by qPCR, locomotor activity assays, differential expression via RNA-seq, and statistical analyses were performed to fulfill their objective. They found that the expression pattern of *atad3* is highly similar to the human paralogs *ATAD3A* AND *ATAD3B* in terms of proteins and domains. RNA-seq expression levels were detected strongly in specific regions of the brain and in the eyes, and since these organs/tissues are also the most affected by *ATAD3A*-associated disorders in humans, they generated knockout models in zebrafish through CRISPR/Cas9. They found that this knockout leads to morphological defects, and the mutant phenotype closely correlates with human phenotype described for *ATAD3A*-related disease. Notable features include microcephaly, small eyes, pericardial edema, and thinning of musculature in homozygous mutants. Additionally, *atad3*-null larvae have decreased mtDNA content, ATP consent, and locomotor activity. Significant differential expression of specific biological pathways such as oxidative phosphorylation, fatty acid metabolism, biosynthesis, etc. was found. Therefore, this study harnessed the advantages of zebrafish as a model animal and opened avenues for investigating therapeutic interventions and further exploring the underlying mechanisms of *ATAD3A*-related disorders (Ezer et al., 2025).

Another way that zebrafish models are useful is that they offer insight into mechanisms of specific disease. Fabry disease was explored in "Proteomic analysis unveils Gb3-independent alterations and mitochondrial dysfunction in a gla$^{-/-}$ zebrafish model of Fabry disease" by Elsaid et al. in zebrafish. Fabry disease (FD) is a rare lysosomal storage disorder caused by mutations in the GLA gene, resulting in reduced or lack of α-galactosidase A activity. This results in an accumulation of globotriaosylceramide (Gb3) and other glycosphingolipids in lysosomes causing cellular impairment and organ failure. As zebrafish naturally lack A4GALT gene and cannot synthesize Gb3, they utilized the zebrafish *gla*$^{-/-}$ mutant model. They obtained the kidney samples from both wild-type and mutant zebrafish and conducted proteome profiling using target mass spectrometry. Additionally, the mitochondria morphology and cristae morphology were examined using electron microscopy. To assess oxidative stress, they measured total antioxidant activity, and performed immunohistochemistry on kidney samples to validate specific proteins.

Their subsequent proteomics analysis of renal tissues from zebrafish revealed the downregulation of lysosomes and mitochondrial-related proteins in the mutant renal tissues, while energy-related pathways including carbon, glycolysis and galactose metabolisms were disturbed. There was also an observed abnormal mitochondrial shape, disrupted cristae morphology, altered mitochondrial volume and lower antioxidant activity in mutant zebrafish. The paper concluded that those results suggest that the alterations observed resemble well-known GLA mutation-related alterations in humans. Importantly, they also revealed novel Gb3-independent pathogenic mechanisms in FD, and understanding of these mechanisms could potentially lead to development of innovative drug screening approaches. The findings illuminate a way for identifying new clinical targets, and offer new therapeutic interventions in Fabry disease (Elsaid et al., 2023).

Discussion:

As mentioned, the zebrafish represents a powerful vertebrate model for investigating mitochondrial disease due to its genetic tractability, rapid development, and high degree of conservation of mitochondrial biology with humans. Key mitochondrial processes like oxidative phosphorylation, cardiolipin remodeling, and mitophagy are preserved between zebrafish and mammals, allowing disease relevant phenotypes to be modeled in vivo. When paired with RNA-sequencing, zebrafish enable organism-level transcriptomic profiling of mitochondrial dysfunction across developmental stages or genetic perturbations. RNA-seq is particularly advantageous in this system because it captures both nuclear-encoded mitochondrial genes and mitochondria-related stress responses, facilitating the identification of mito-nuclear imbalance, compensatory metabolic pathways, and downstream signaling effects. Additionally, the small size and high fecundity of zebrafish allow for biologically replicated RNA-seq experiments at low cost, increasing statistical power and experimental reproducibility.

Despite these advantages, there are important limitations to both the zebrafish model and RNA-seq that should be considered when interpreting mitochondrial data. Zebrafish larvae RNA-seq experiments often rely on whole-organism or pooled samples, which can obscure tissue-specific mitochondrial phenotypes, particularly in metabolically specialized tissues such as heart, muscle, and brain. While RNA-seq robustly measures transcript abundance, mitochondrial dysfunction is frequently driven by post-transcriptional mechanisms, including altered protein assembly, enzyme activity, and metabolite availability, which are not captured at the RNA level. Additionally, mitochondrial DNA heteroplasmy, copy number variation, and respiratory chain efficiency cannot be directly inferred from transcriptomic data alone, limiting RNA-seq's ability to fully characterize primary mitochondrial defects.

From an analytical perspective, RNA-seq of mitochondrial pathways presents additional challenges. Mitochondrial transcripts can be underrepresented depending on library preparation methods, and the high sequence similarity between mitochondrial and nuclear pseudogenes (NUMTs) can complicate read alignment. Pathway enrichment analyses also depend heavily on annotation quality and may overemphasize transcriptional compensation rather than functional recovery. As demonstrated in zebrafish tafazzin-deficiency studies, increased expression of oxidative phosphorylation or mitophagy genes may reflect stress responses rather than restored mitochondrial function. Therefore, while RNA-seq is a highly effective discovery tool for identifying mitochondrial-associated transcriptional changes, its conclusions are strongest when integrated with complementary approaches such as proteomics, metabolomics, mitochondrial respiration assays, or targeted mtDNA analyses. Together, these multi-omics strategies allow zebrafish RNA-seq data to be interpreted within a more complete functional framework of mitochondrial disease.

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